

# Reduction formulae

## Key Points

An integration involving an unknown constant  $n$  can usually be found by a reduction formula

Repeated integration by parts allows us to find formulae for integrations of functions involving  $n$

When choosing what to differentiate and integrate, differentiate the part to make  $n$  decrease

## Basic Skills

**Q1.** State the integration by parts formula

**Q2.** Use integration by parts to find the following

a)  $\int x \sin x \, dx$

b)  $\int x^5 \ln 2x \, dx$

c)  $\int x \sqrt{x+2} \, dx$

d)  $\int 4xe^{\frac{1}{2}x} \, dx$

**Q3.** Using integration by parts show

a)  $\int x^2 \sin x \, dx = -x^2 \cos x + \int 2x \cos x \, dx$

b)  $\int 2x \cos x \, dx = 2x \sin x - \int 2 \sin x \, dx$

c) Hence find  $\int x^2 \sin x \, dx$

## Competency Skills

**Q4.** In order to find  $I_n = \int x^n \sin x \, dx$  first

a) Show that  $\int x^n \sin x \, dx = -x^n \cos x + \int nx^{n-1} \cos x \, dx$

b) Then show that  $\int x^{n-1} \cos x \, dx = x^{n-1} \sin x - (n-1) \int x^{n-2} \sin x \, dx$

c) Hence show  $I_n = -x^n \cos x + nx^{n-1} \sin x - n(n-1)I_{n-2}$

**Q5.** By using integration by parts twice find the reduction formulae for  $I_n = \int x^n \cos x \, dx$

**Q6.** By using integration by parts to find the reduction formulae for  $I_n = \int x^n e^x \, dx$

**Q7.** Given that  $I_n = \int_0^k (k-x)^n \sin x \, dx$

a) Show that  $I_n = k^n - n(n-1)I_{n-2}$

b) Hence find  $\int_0^{\frac{3\pi}{2}} \left(\frac{3\pi}{2} - x\right)^2 \sin x \, dx$

## Advanced Skills

**Q8.** Explain why does the integration of  $x^n \ln x$  not require a reduction formulae but  $\ln^n x$  does

**Q9.** Given that  $I_n = \int_0^4 \frac{x^n}{\sqrt{16-x^2}} \, dx$

a) First find an expression for  $\int \frac{x}{\sqrt{16-x^2}} \, dx, 0 \leq x \leq 4$

b) Next show that  $I_n = \frac{16(n-1)}{n} I_{n-2}$  for  $n \geq 2$

c) Hence find  $I_5$

**Q10.** Find the reduction formula for  $I_n = \int_0^{\ln 3} \tanh^n x \, dx$

## Extension

**Q11.** Find the reduction formula for  $I_n = \int e^x \cos^n x \, dx$

Hint: differentiate the trig functions in each iteration

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